

EMILY G. POTTEBAUM

Curriculum Vitae

Wright Laboratory
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EDUCATION

Yale University

Doctorate of Philosophy in Physics 2022 - Present
Masters of Science and Philosophy in Physics December 2025

Iowa State University

Bachelor of Science in Physics May 2022

RESEARCH EXPERIENCE

Graduate Research Fellow

Advisor: Helen Caines

Yale University
August 2024 - Present

- STAR experiment: Lund jet plane measurement in p+p collisions (ongoing), detector operator shifts
- EIC/ePIC experiment: participation in and analysis software development for LFHCal test beam campaigns (ongoing), SiPM QA testing for LFHCal (ongoing), fixed target collision simulations

Graduate Research Assistant

Advisor: Karsten Heeger

Yale University
July 2022 - August 2024

- Hardware, software, and electronics development of a muon veto system for the CUORE & CUPID experiments
- Supervised instrumentation R&D work conducted by undergraduates

Undergraduate Research Assistant

Advisor: Mayly Sánchez

Iowa State University
May 2020 - June 2022

- Data acquisition and analysis for transparency monitoring of the ANNIE neutrino experiment's water Cherenkov detector
- Analysis of beam-correlated neutron background rates for ANNIE Phase II

Undergraduate Research Assistant

Advisor: Marzia Rosati

Iowa State University
January 2019 - May 2022

- Upgraded jet reconstruction software written for multi-version compatibility with ROOT, PYTHIA
- Study on projected J/ψ production in jets for sPHENIX

NSF REU Summer Intern

Advisor: David Stuart

University of California, Santa Barbara
June 2021 - August 2021

- Designed and tested a summing amplifier PCB for the milliQan experiment slab detector
- Developed a peak detection and fitting algorithm for PMT calibration

TEACHING EXPERIENCE

Graduate Teaching Fellow

- PHYS 1700, "Nuclear Physics: The Good, the Bad, and the Misunderstood"
- PHYS 4450L, "Advanced Physics Laboratory"

Yale University
Fall 2025
Spring 2024

· PHYS 2060L, “Modern Physical Measurement” Fall 2022 - Fall 2023, Fall 2024

Undergraduate teaching

Iowa State University

- Peer Mentor — LDST 122 (now 1220), “Leading with Purpose” Fall 2020
- Academic Success Center Tutor — PHYS 121 (now 1310), “General Physics I” Spring 2019

OUTREACH AND COMMUNITY ENGAGEMENT

Pathways to Science Summer Scholars Program

Yale University

- “Physics Demonstrations” co-developer and co-leader — [Science communication workshop](#) 2025
- “Mini YPO” co-leader — [Enrichment activity](#) 2025
- “Let’s Smash some Protons” volunteer — [Physics workshop](#) 2025
- “RHIGscape” volunteer — [Enrichment activity](#) 2025
- “Discover the Invisible Universe” co-leader — [Enrichment activity](#) 2023
- “Cool Physics” leader — [Enrichment activity](#) 2022

Girls’ Science Investigations

Yale University

- Lead developer and co-presenter of demo shows
([1](#), [2](#), [3](#), [4](#), [5](#), [6](#)) 2023 - Present

Yale Physics Olympics

Yale University

- Lead developer and co-presenter of demo shows ([1](#), [2](#)) 2025 - Present
- Organizing Committee member 2023 - Present
- Event lead ([1](#)) 2024
- Event volunteer ([1](#)) 2023

Haunted Hallway and Halloween Demo Show

Yale University

- Lead developer and co-presenter of the demo show ([1](#), [2](#), [3](#)) 2023 - Present
- Organizing Committee founding and current member 2023 - Present

New Haven Public Schools

New Haven, CT

- Pathways club guest speaker — Hill Regional Career High School December 2025
- Tabletop demos — Title I Parent Night October 2025
- Tabletop demos — New Haven Science Fair June 2025
- Science fair judge — Edgewood Magnet School March 2025

Gender Representation in Physics

Yale University

Graduate student organization

- Co-chair 2025 - Present
- Administrator 2023 - Present
- First year liaison 2022 - 2023

QuARK — Queer in Physics

Yale University

Student organization

- Administrator 2025 - Present
- Leadership team 2023 - Present

Yale Community

Yale University

- Local 33 graduate student union organizer

2022 - Present

Miscellaneous

- Feature in [Amazing Iowa Scientists](#) book by Katy Swalwell Published April 2026
- Global Citizenship Day [guest speaker](#) — Scarsdale High School October 2025
- Nuclear Physics on the Hill day for science funding advocacy in DC April 2025
- “Invisible Universe” [demos](#) — Upward Bound Math Science program, Waterbury May 2023

Society of Women in Physics and Astronomy

Iowa State University

Student organization

- Vice President 2021 - 2022
- President 2019 - 2021

Physics and Astronomy Club

Iowa State University

Student organization

- Secretary 2020 - 2022
- Outreach volunteer at regional schools, planetarium shows, local retirement home 2018 - 2022

HONORS AND AWARDS

D. Allan Bromley Fellowship	2025 - 2026
Nathan Hale Associates Fellowship Award	2025 - 2026
NSF GRFP — Honorable Mention	2024
Phi Beta Kappa Honor Society	2021
Gordon C. Danielson Memorial Endowment Scholarship	2021
Bernice Black Durand Undergraduate Research Scholarship	2020
David W. Collins Memorial Scholarship	2019
Physics Achievement Award for Superior Academic Performance	2019

TALKS AND POSTERS

Measurement of the Lund Jet Plane in 200 GeV pp collisions with STAR	June 2026
Conference on Hard and Electromagnetic Probes of High-Energy Nuclear Collisions (poster)	
The ePIC Longitudinally Segmented Forward HCal	June 2026
CALOR 2026 (talk)	
My Pathway to Nuclear Physics	January 2026
Hill Regional Career High School (invited talk)	
Jet Physics with the STAR Forward Upgrade	June 2025
Hard Probes in Proton and Nucleus Collisions — CTU-Yale workshop (talk)	
Gender Representation in Physics	October 2025
Scarsdale High School Global Citizenship Day (invited talk with Morgan Knuesel)	
Searching for Hidden Matter with milliQan	January 2022
Conference for Undergraduate Women in Physics (poster)	
ANNIE Calibration and Analysis in Phase II	June 2022
International Conference on Neutrino Physics and Astrophysics (contributing author on poster)	

PUBLICATIONS

Primary author

- *Performance of a SiPM-based, plastic scintillator muon veto prototype for CUPID.* 8 Aug 2025. Published in [JINST 20 P08020](#)

STAR

- *Probing late-stage hadronic interactions at high baryon density via K^{*0} production in the RHIC beam energy scan program.* May 2026. Published in [Phys. Lett. B 876 \(2026\) 140389](#).
- *Observation of charmonium sequential suppression in heavy-ion collisions at the Relativistic Heavy Ion Collider.* 24 Mar 2026. Published in [Phys. Rev. Lett. 136, 122302 \(2026\)](#).
- *Energy independence of the Collins asymmetry in $p\uparrow p$ collisions.* 26 Dec 2025. Published in [Phys. Rev. Lett. 135, 261902](#).
- *Onset of constituent quark number scaling in heavy-ion collisions at RHIC.* 11 Aug 2025. Published in [Phys. Rev. Lett. 135, 072301](#).

CUORE/CUPID

- *End-to-end data analysis methods for the CUORE experiment.* 2026. Submitted to Eur. Phys. J. C [[arXiv:2510.25720](#)]
- *Sensitivity of the CUPID experiment to $0\nu\beta\beta$ decay of ^{100}Mo .* 2 Feb 2026. Accepted for publication in Eur. Phys. J. C [[arXiv:2504.14369](#)]
- *The detection of marine microseismic activity with the CUORE tonne-scale cryogenic experiment.* 26 Feb 2026. Published in [Communication Physics 9, 121](#)
- *Exploring the keV-scale physics potential of CUORE.* 23 Jan 2026. Published in [Phys. Rev. D 113, 012012](#)
- *Constraints on lepton number violation with the 2 tonne-year CUORE dataset.* 16 Oct 2025. Published in [Science 390, 6777, 1029-1032](#)
- *A gravity-based mounting approach for large-scale cryogenic calorimeter arrays.* 2 Sep 2025. Published in [Eur. Phys. J. C](#)
- *Half-life and precision shape measurement of $2\nu\beta\beta$ decay of ^{130}Te .* 22 Aug 2025. Published in [Phys. Rev. Lett. 135, 082501](#)
- *CUPID, the CUORE upgrade with particle identification.* 3 July 2025. Published in [Eur. Phys. J. C](#)
- *Reconstruction of cosmic-ray muon events with CUORE.* 2025. Submitted to Eur. Phys. J. C [[arXiv:2509.05528](#)]
- *Search for fractionally charged particles with CUORE.* 12 Dec 2024. Published in [Phys. Rev. Lett. 133, 241801](#)
- *Data-driven background model for the CUORE experiment.* 11 Sep 2024. Published in [Phys. Rev. D 110, 052003](#)

ANNIE

- *Reconstruction of neutrino events in the Accelerator Neutrino Neutron Interaction Experiment.* 23 Mar 2026. Published in [JINST 21 P03039](#)
- *First light from beam neutrinos on an LAPPD in ANNIE.* 3 Feb 2026. Published in [JINST 21 P02002](#)